

GRACE N. SMITH



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scan for clickable links!

Education

Massachusetts Institute of Technology

Sep. 2019 – May 2026

Candidate for Bachelors of Science: Mechanical Engineering

Cambridge, MA

Magnolia West High School

Aug. 2015 – May 2019

AP Scholar with Distinction, 15 AP Courses | GPA: 4.29

Magnolia, TX

Relevant Coursework

- Nonlinear Dynamics
- Dynamics and Controls I, II
- Measurement and Instrumentation
- Thermodynamics of Biomolecular Systems
- Electronic Musical Instrument Design
- Engineering Systems Design
- Electronics for Mechanical Systems
- Numerical Analysis
- Social Entrepreneurship
- Making Immersive Experiences in VR

Research Experience

Undergraduate Researcher

October 2022 – December 2023

Mentored by Teemu Miettinen | MIT Koch Institute

Cambridge, MA

- Developing an automation for the analysis of cell cortex thickness from fluorescent imaging data, for investigation into the nature of cell growth
- Culminated in a paper submitted to PNAS

Work Experience

Design and Manufacturing Consultant

January 2024 - August 2024

SpectroArt

Remote

- Advised on SLA printing techniques to ensure startup success on first few client deals
- Designed workflow solutions for generating 3D models compatible with manufacturing constraints
- Provided assistance and training in use of specialized 3D-modeling software

Design Engineering Intern

Fall 2023

Transaera

Somerville, MA

- Worked on a fast-paced engineering team to prototype a portable AC unit with 40% energy efficiency gains
- Introduced engineering team to new 3D-printing technologies and techniques, saving weeks on lead times for previously-outsourced parts
- Cultivated reputation as innovative problem-solver, regularly consulted by colleagues at all levels to workshop complex engineering and manufacturing solutions

Mechanical Engineering Tutor

Summer 2023

MIT Women's Technology Program | MIT Mechanical Engineering

Cambridge, MA

- Worked on a team to deliver a world-class engineering program to 50 high-school students
- Led engaging classes in brainstorming, design, and rapid-prototyping/manufacturing
- Returned in subsequent years to assist and advise in manufacturing-related activities

Software Engineer

October 2020 - January 2021

Uzephra

Remote

- Contributed to the construction of a travel web application as a back-end software engineer
- Developed a COVID-19 risk model for disease spread on airplanes based on the Wells-Riley model and extended to consider cabin-to-cabin airflow as a factor in virus spread

Technical Skills

Programming Languages and Frameworks: Julia, Python, MATLAB, MAX/MSP, Java, R, C/C++, Arduino, Excel, PyTorch, comfortable in Linux-based systems

Languages: English (native), Russian (native), Spanish (basic/conversational), Japanese (basic/conversational)

Other: Physical and digital fabrication, Fusion360, SOLIDWORKS, Onshape, Adobe Creative Suite, Blender